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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

Continuous Time Signal Laplace Transform

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

35	27	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	13 questions
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Answer key	at the end

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Marks 1 13 questions

Q1. GATE ECE 2026

The Laplace Transform of the signal $x(t) = u(t-2) * (tu(t))$ is given by which of the following expressions?

["*" represents convolution operator]

(A) $\frac{e^{-2s}}{s^2(s-2)}$ (B) $\frac{e^{-2(s-2)}}{s^3}$ (C) $\frac{se^{-2s}}{(s-2)^2}$ (D) $\frac{e^{-2s}}{s^3}$

Open solution

Q2. GATE ECE 2015 Set 2

The bilateral Laplace transform of a function $f(t) = \begin{cases} 1 \, if \, a \le t \le b \\ 0 \, otherwise \end{cases}$ is

(A) $\frac{a-b}{s}$ (B) $\frac{e^s(a-b)}{s}$

(C) $\frac{e^{-as}-e^{-bs}}{s}$ (D) $\frac{e^{s(a-b)}}{s}$

Open solution

Q3. GATE ECE 2014 Set 1

A continuous, linear time - invariant filter has an impulse response h(t) described by

$h(t) = \begin{cases} 3 \, for \, 0 \le t \le 3 \\ 0 \, otherwise \end{cases}$

When a constant input of value 5 is applied to this filter, the steady state output is ____.

Open solution

Q4. GATE ECE 2014 Set 3

The input $-3e^{2t} \, u(t)$, where u(t) is the unit step function, is applied to a system with transfer function $\frac{s-2}{s+3}$. If the initial value of the output is -2, then the value of the output at steady state is _____.

Open solution

Q5. GATE ECE 2007

If the Laplace transform of a signal y(t) is $Y(s) = \frac{1}{s(s-1)}$, then its final value is

(A) -1 (B) 0 (C) 1 (D) Unbounded

Open solution

Q6. GATE ECE 2003

The Laplace transform of i(t) tends to

$I(s) = \frac{2}{s(1+s)}$

As $t \to \infty$, the value of i(t) tends to

(A) 0 (B) 1 (C) 2 (D) ∞

Open solution

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Q7. GATE ECE 2000

Open solution

Given that $L[f(t)] = \frac{s+2}{s^2+1}$,

$$L[g(t)] = \frac{s^2+1}{(s+3)(s+2)},$$

$$h(t) = \int_0^t f(\tau)\, g(t-\tau)\, d\tau,$$

$L[h(t)]$ is

(A) $\frac{s^2+1}{s+3}$

(B) $\frac{1}{s+3}$

(C) $\frac{s^2+1}{(s+3)(s+2)} + \frac{s+2}{s^2+1}$

(D) None of the above

Q8. GATE ECE 1999

Open solution

If $L[f(t)] = F(s)$, then $L[f(t-T)]$ is equal to

(A) $e^{sT}F(s)$

(B) $e^{-sT}F(s)$

(C) $\frac{F(s)}{1+e^{sT}}$

(D) $\frac{F(s)}{1-e^{-sT}}$

Q9. GATE ECE 1998

Open solution

If L$[f(t)] = \omega/(s^2+\omega^2)$, then the value of $\underset{t \to \infty}{Lim}\, f(t)$

(A) cannot be determined

(B) is zero

(C) is unity

(D) is infinite

Q10. GATE ECE 1997

Open solution

The Laplace Transform of $e^{\alpha t}.\cos(\alpha t).u(t)$ is equal to

(A) $\frac{(s-\alpha)}{(s-\alpha)^2+\alpha^2}$

(B) $\frac{(s+\alpha)}{(s+\alpha)^2+\alpha^2}$

(C) $\frac{1}{(s-\alpha)^2}$

(D) none of the above

Q11. GATE ECE 1995

Open solution

If L$[f(t)] = \frac{2(s+1)}{s^2+2s+5}$, then $f(0+)$ and $f(\infty)$ are given by

(A) 0, 2 respectively

(B) 2, 0 respectively

(C) 0, 1 respectively

(D) 2/5, 0 respectively

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Q12. GATE ECE 1995

The final value theorem is used to find the

(A) steady state value of the system output
(B) initial value of the system output
(C) transient behavior of the system output
(D) none of these

Open solution

Q13. GATE ECE 1994

The laplace transform of a unit ramp function starting at t=a, is

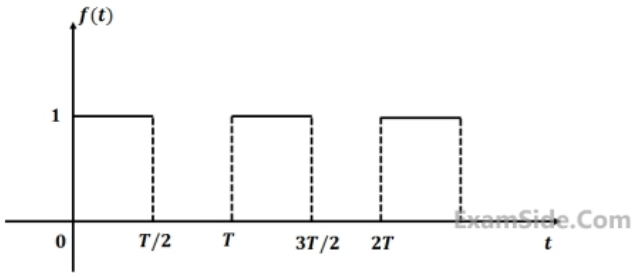
(A) $\frac{1}{(s+a)^2}$
(B) $\frac{e^{-as}}{(s+a)^2}$
(C) $\frac{e^{-as}}{s^2}$
(D) $\frac{a}{s^2}$

Open solution

Marks 2 20 questions

Q14. GATE ECE 2016 Set 1

The Laplace transform of the causal periodic square wave of period T shown in the figure below is



(A) $F(s) = \frac{1}{1+e^{-sT/2}}$
(B) $F(s) = \frac{1}{s\left[1+e^{\frac{sT}{2}}\right]}$
(C) $F(s) = \frac{1}{s(1+e^{-sT})}$
(D) $F(s) = \frac{1}{1-e^{-sT}}$

Open solution

Q15. GATE ECE 2015 Set 2

Let x(t) = a s(t) +s(-t) with s(t) = $\beta e^{-4t}u(t)$, where u(t) is unit step function. If the bilateral Laplace transform of x(t) is

$$X(S) = \frac{16}{S^2 - 16} \quad -4 < \mathrm{Re}\{s\} < 4;$$

Then the value of β is _____.

Open solution

Q16. GATE ECE 2015 Set 1

The solution of the differential equation $\frac{h(t+1)}{h(t)}$ $\frac{d^2y}{dt^{\to 2}} + \frac{2\,dy}{dt} + y = 0$ with $y(0) = y'(0) = 1$ is

(A) $(2-t)e^t$
(B) $(1+2t)e^{-t}$
(C) $(2+t)e^{-t}$
(D) $(1-2t)e^t$

Open solution

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Q17. GATE ECE 2014 Set 1

A system is described by the following differential equation, where u(t) is the input to the system and y(t) is output of the system $\dot{y}(t) + 5y(t) = u(t)$
When y(0) = 1 and u(t) is a unit step function, y(t) is

(A) $0.2+0.8e^{-5t}$ (B) $0.2-0.2e^{-5t}$ (C) $0.8+0.2e^{-5t}$ (D) $0.8-0.8e^{-5t}$

Open solution

Q18. GATE ECE 2014 Set 4

The unilateral Laplace transform of F(t) is $\frac{1}{s^2+s+1}$. Which one of the following is the unilateral Laplace transform of g(t) = $t \cdot f(t)$

(A) $\frac{-s}{(s^2+s+1)^2}$ (B) $\frac{-(2s+1)}{(s^2+s+1)^2}$

(C) $\frac{s}{(s^2+s+1)^2}$ (D) $\frac{2s+1}{(s^2+s+1)^2}$

Open solution

Q19. GATE ECE 2014 Set 4

A stable linear time invariant (LTI) system has a transfer function H(s) = $\frac{1}{s^2+s-6}$. To make this system casual it needs to be cascaded with another LTI system having a transfer function $H_1(s)$. A correct choice for $H_1(s)$ among the following options is

(A) s + 3 (B) s - 2 (C) s - 6 (D) s + 1

Open solution

Q20. GATE ECE 2014 Set 4

A casual LTI system has zero initial conditions and impulse response h(t). Its input x(t) and output y(t) are related through the linear constant - coefficient differential equation

$$\frac{d^2y(t)}{dt^2} + \alpha\frac{dy(t)}{dt} + \alpha^2 y(t) = x(t).$$

Let another signal g(t) be defined as $(t) = \alpha^2 \int_0^t h(\tau)d\tau + \frac{dh(t)}{dt} + \alpha h(t)$.

If G(s) is the Laplace transform of g(t), then the number of poles of G(s) is _____.

Open solution

Q21. GATE ECE 2014 Set 3

Let h(t) denote the impulse response of a casual system with transfer function $\frac{1}{s+1}$. Consider the following three statements.
S1: The system is stable.
S2: $\frac{h(t+1)}{h(t)}$ is independent of t for t > 0.
S3: A non-casual system with the same transfer function is stable.

For the above system,

(A) only S1 and S2 are true (B) only S2 and S3 are true

(C) only S1 and S3 are true (D) S1, S2 and S3 are true

Open solution

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Q22. GATE ECE 2013

A system is described by the differential equation

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y(t) = x(t)$$

Open solution

Let x(t) be a rectangular pulse given by

$$x(t) = \begin{cases} 1 & 0 \le t \le 2 \\ 0 & otherwise \end{cases}$$

Assuming that y(0) = 0 $\frac{dy}{dt} = 0$ at t = 0, the Laplace transform of y(t) is

(A) $\frac{e^{-2s}}{s(s+2)(s+3)}$

(B) $\frac{1-e^{-2s}}{s(s+2)(s+3)}$

(C) $\frac{e^{-2s}}{(s+2)(s+3)}$

(D) $\frac{1-e^{-2s}}{(s+2)(s+3)}$

Q23. GATE ECE 2011

If $F(s) = L[f(t)] = \frac{2(s+1)}{s^2+4s+7}$ then the initial and final values of f(t) are respectively

(A) 0, 2 (B) 2, 0 (C) 0, 2/7 (D) 2/7, 0

Open solution

Q24. GATE ECE 2010

Given f(t) = $L^{-1}\left[\frac{3s+1}{s^3+4s^2+(K-3)s}\right]$. If $\underset{t \to \infty}{Lim} f(t) = 1,$ then the value of K is

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q25. GATE ECE 2009

Given that F(s) is the one-sided Laplace transform of f(t), the Laplace transform of $\int_0^t f(\tau)\,d\tau$ is

(A) $sF(s) - f(0)$

(B) $\frac{1}{s}F(s)$

(C) $\int_0^s F(\tau)d\tau$

(D) $\frac{1}{s}[F(s) - f(0)]$

Open solution

Q26. GATE ECE 2006

Consider the function f(t) having Laplace transform $F(s) = \frac{\omega_0}{s^2+\omega_0{}^2} \quad \mathrm{Re}(s) > 0$
The final value of f(t) would be:

(A) 0 (B) 1

(C) $-1 \le f(\infty) \le 1$ (D) ∞

Open solution

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Q27. GATE ECE 2005

In what range should Re(s) remain so that the Laplace transform of the function $e^{(a+2)t+5}$ exists?

(A) Re(s) > a+2 (B) Re(s) > a+7 (C) Re(s) < 2 (D) Re(s) > a+5

Open solution

Q28. GATE ECE 2002

The Laplace transform of a continuous - time signal x(t) is $X(s) = \frac{5-s}{s^2-s-2}$. If the Fourier transform of tyhis signal exists, then x(t) is

(A) $e^{2t}u(t) - 2e^{-t}u(t)$ (B) $-e^{2t}u(-t) + 2e^{-t}u(t)$

(C) $-e^{2t}u(-t) - 2e^{-t}u(t)$ (D) $e^{2t}u(-t) - 2e^{-t}u(t)$

Open solution

Q29. GATE ECE 1996

The inverse Laplace transform of the function $\frac{s+5}{(s+1)(s+3)}$ is

(A) $2e^{-t} - e^{\to -3t}$ (B) $2e^{-t} + e^{\to -3t}$

(C) $e^{-t} - 2e^{-3t}$ (D) $e^{-t} + 2e^{-3t}$

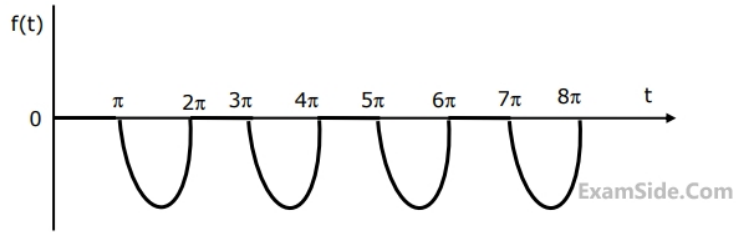
Open solution

Q30. GATE ECE 1993

The Laplace transform of the periodioc function f(t) describe4d by the curve below, i.e.,

$$f(t) = \begin{cases} \sin t \ \ if \ (2n-1)\pi \le t \le 2n\pi \\ 0 \quad otherwise \end{cases}$$

is _____. (fill in the blank), n is an integer.



Open solution

Q31. GATE ECE 1993

If $F(s) = L[f(t)] = \frac{K}{(s+1)(s^2+4)}$ then $\underset{t \to \infty}{Lim} f(t)$ is given by

(A) K/4 (B) zero (C) infinite (D) undefined

Open solution

Q32. GATE ECE 1988

The Laplace transform of a function f(t)u(t), where f(t) is periodic with period T, is A(s) times the Laplace transform of its first period. Then

(A) A(s) = s (B) A(s) = 1/(1-exp(-Ts))

(C) A(s) = 1/(1+exp(-Ts)) (D) A(s) = exp (Ts)

Open solution

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Q33. GATE ECE 1987

Laplace transform of the functions t u(t) and u(t) sin(t) are respectively:

(A) $\frac{1}{s^2}, \frac{s}{s^2+1}$

(B) $\frac{1}{s}, \frac{1}{s^2+1}$

(C) $\frac{1}{s^2}, \frac{1}{s^2+1}$

(D) $s, \frac{s}{s^2+1}$

Open solution

Marks 5 2 questions

Q34. GATE ECE 1995

A sinsoidal signal, v(t) = A sin(t), is applied to an ideal full-wave rectifier. Show that the Laplace Transform of the output can be written in the form, $V_0(s) = \frac{A}{s^2+1} Coth(\alpha s)$, where α is a constant. Determine the value of α.

Open solution

Q35. GATE ECE 1991

Find the Laplace transform of the waveform x(t) shown in Fig.1.

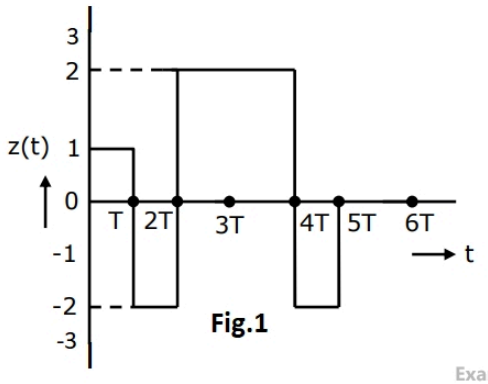


Fig.1

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Open solution

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Answer Key

Marks 1

1. **D**	2. **C**	3. **45**	4. **-0.01 to 0.01**	5. **D**
6. **C**	7. **B**	8. **B**	9. **A**	10. **A**
11. **B**	12. **A**	13. **C**		

Marks 2

14. **C**	15. **-2**	16. **B**	17. **A**	18. **D**
19. **B**	20. **1**	21. **A**	22. **B**	23. **B**
24. **D**	25. **B**	26. **C**	27. **A**	28. **C**
29. **A**	30. $F(s) = -\frac{e^{-\Pi S}}{(s^2+1)(1-e^{-\Pi S})}$	31. **D**	32. **B**	33. **C**

Marks 5

34. $\alpha = \frac{\pi}{2}$	35. **Laplace transform** = $\frac{1}{5}\left[1 - 3e^{-Ts} + 4e^{-2Ts} - 4e^{-4Ts} + 2e^{-5Ts}\right]$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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